

DAY 172/365 • PART 3 OF 3

Build a tree by hand

A full worked example, plus how to stop it from overfitting.

The data: 8 customers

Will a customer buy? We know two things about each one:

#	Bought before?	Age > 30?	Buys?
1	Yes	Yes	Buy
2	Yes	Yes	Buy
3	Yes	Yes	Buy
4	Yes	No	Don't
5	No	Yes	Don't
6	No	Yes	Don't
7	No	Yes	Don't
8	No	No	Don't

A made-up dataset for learning.

Step 1: How messy is everyone together?

At the root, we have 3 buyers and 5 non-buyers.

$$\begin{aligned}\text{Entropy} &= - (3/8) \log_2(3/8) - (5/8) \\ &\log_2(5/8) \\ &= 0.954\end{aligned}$$

That's close to the maximum of 1, so the group is quite mixed. Our job is to ask questions that push this number down.

Remember from Part 2: **information gain = entropy before - weighted entropy after.**

Step 2: Test both questions

Bought before?

Yes: 4 customers (3 buy, 1 doesn't), entropy 0.81

No: 4 customers (all don't), entropy 0

$$\text{Weighted} = \frac{1}{2} \times 0.81 + \frac{1}{2} \times 0 = 0.41$$

Age > 30?

Yes: 6 customers (3 buy, 3 don't), entropy 1.0

No: 2 customers (all don't), entropy 0

$$\text{Weighted} = \frac{3}{4} \times 1.0 + \frac{1}{4} \times 0 = 0.75$$

Question	Information gain
Bought before?	$0.954 - 0.41 = 0.55$
Age > 30?	$0.954 - 0.75 = 0.20$

Bought before? wins, so it becomes the root.

Step 3: Split, then repeat

“No” branch: customers 5 to 8 all don’t buy.
Entropy 0, so the group is pure. Make it a leaf:
Don’t buy.

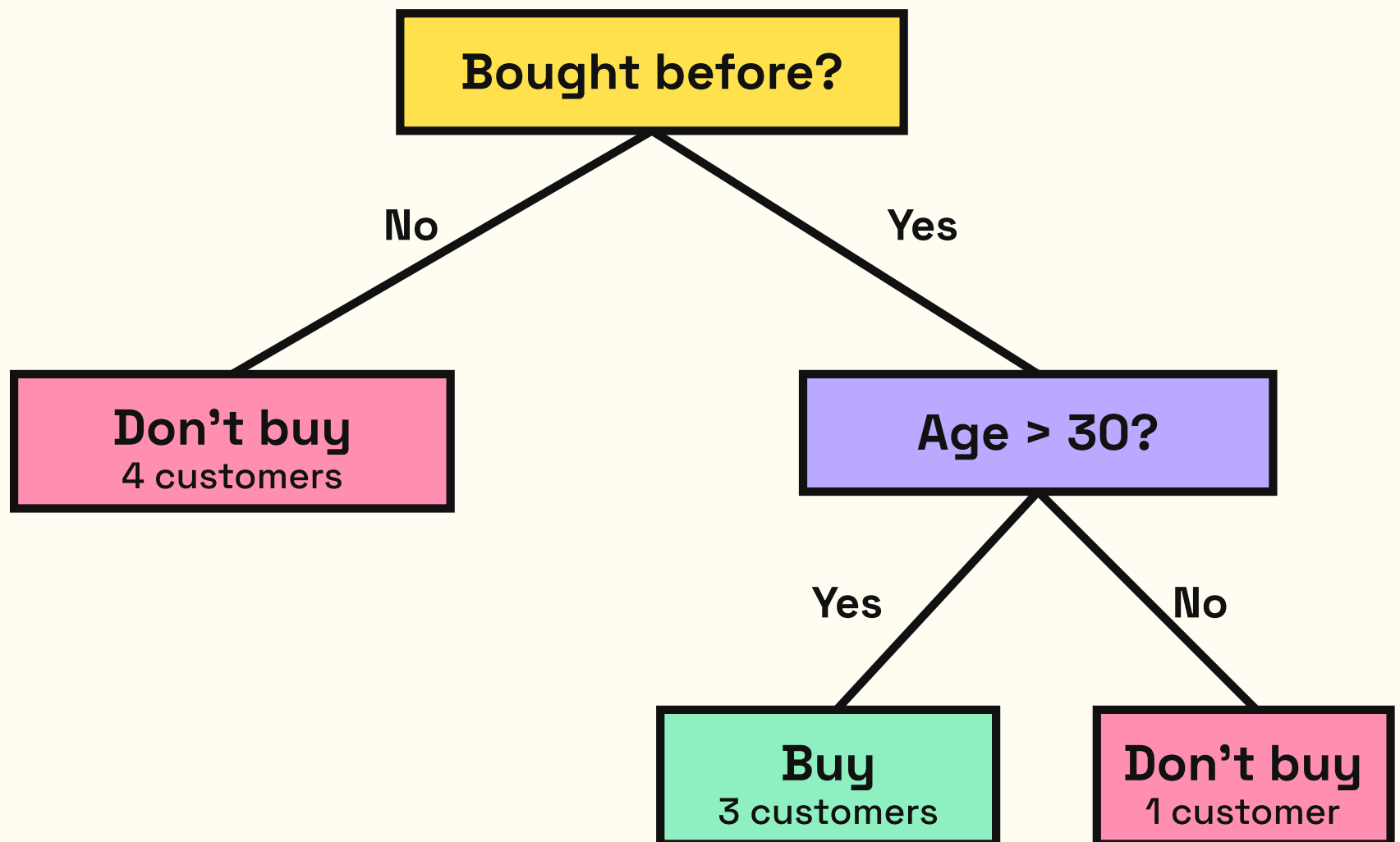
“Yes” branch: customers 1 to 4 are still mixed (3 buy, 1 doesn’t). Ask another question.

Only one question is left: **Age > 30?**

Age > 30?	Who	Result
Yes	Customers 1, 2, 3	All buy
No	Customer 4	Doesn’t buy

Both groups are pure, so the tree is finished.

The finished tree



Read it as rules: no past purchase → don't buy. Bought before and over 30 → buy. It gets all 8 training customers right.

The catch: perfect on 8 rows isn't proof

On real data, a tree left to grow until every group is pure will memorize noise. It might add a special leaf for one odd customer who bought a gift once. That's Student B again, from my Day 168 post.

Control	What it does
<code>max_depth</code>	Limits how many questions in a row, e.g. 3
<code>min_samples_leaf</code>	Every leaf needs at least N customers, e.g. 5
Pruning	Grow big, then cut branches that don't help on unseen data

```
tree = DecisionTreeClassifier(  
    max_depth=3, min_samples_leaf=5)
```

Pick these values with cross-validation, as I did for alpha.

THE WHOLE SERIES

Decision trees in 3 lines

Part 1: a flowchart that learns its own questions.
Predict by walking root to leaf.

Part 2: it picks the question with the highest
information gain (or lowest Gini).

Part 3: repeat until groups are pure, but limit
depth so it doesn't memorize.

**Which would you rather explain to a manager: a tree
or a logistic regression?**